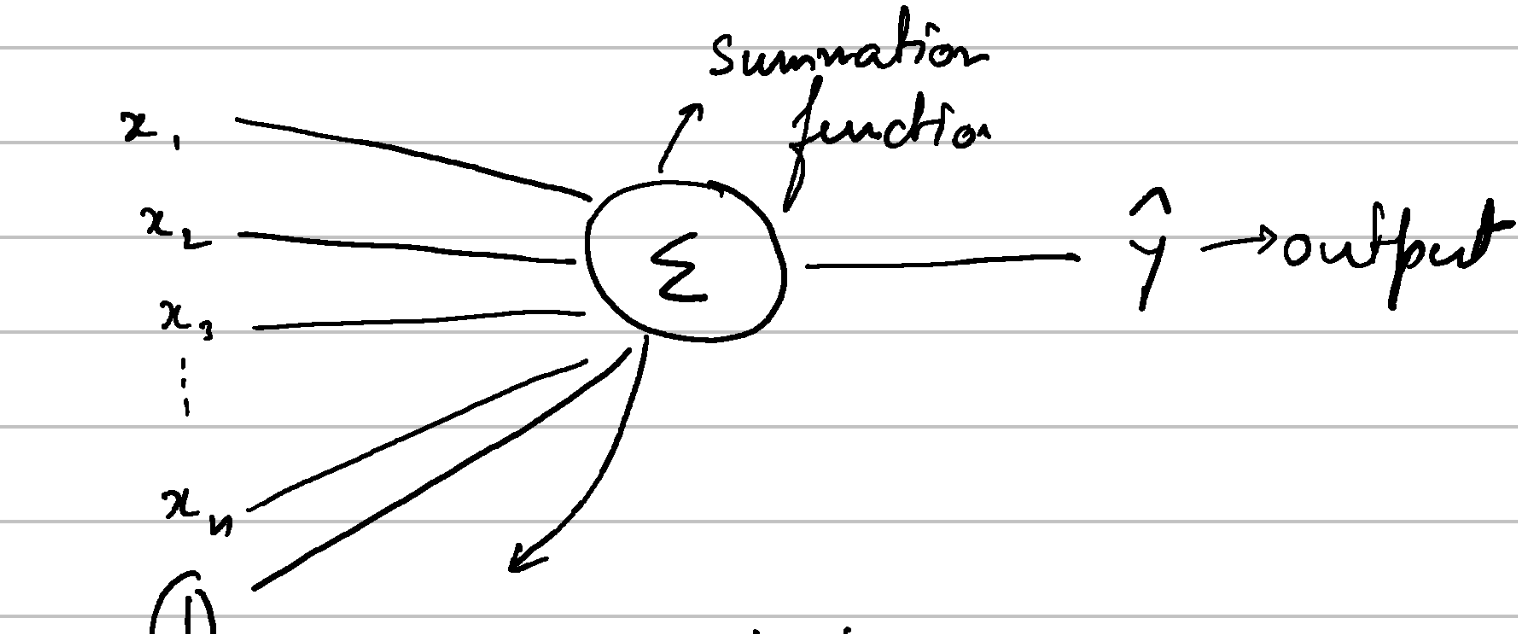
# Lesson 5

## Perceptron

Inputs




each feature/input will have a corresponding weight attached to it

$w_1, w_2 \ldots\ldots w_n$

$$\hat{y} = w_1 x_1 + w_2 x_2 \dots w_n x_n + \underset{\text{bias}}{w_0}$$

⇒ The goal is to now find optimal w's that will make accurate predictions.

⇒ Loss function → The function that calculates the error between a true output & model's prediction.

eg Mean squared loss → $\frac{1}{2}(y-\hat{y})^2$

# Optimization using gradient descent

$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

$b \rightarrow$ bias weight

$$L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$$

$$\Rightarrow \quad w_{i+1} \rightarrow w_i - \alpha \frac{\partial L}{\partial w_i} \rightarrow \text{gradient.}$$

$w_i \downarrow$ initial start value

$$W_i = W_i - \alpha \nabla f$$

For the given example →

$$\frac{\partial L}{\partial b} = (y - \hat{y}) \cdot (-1) \cdot \frac{d\hat{y}}{d_b}$$

$$= (\hat{y} - y) \cdot (1) = (w_1 x_1 + w_2 x_2 + b - y)$$

$$\frac{\partial L}{\partial w_1} = (w_1 x_1 + w_2 x_2 + b - y) \cdot w_1$$

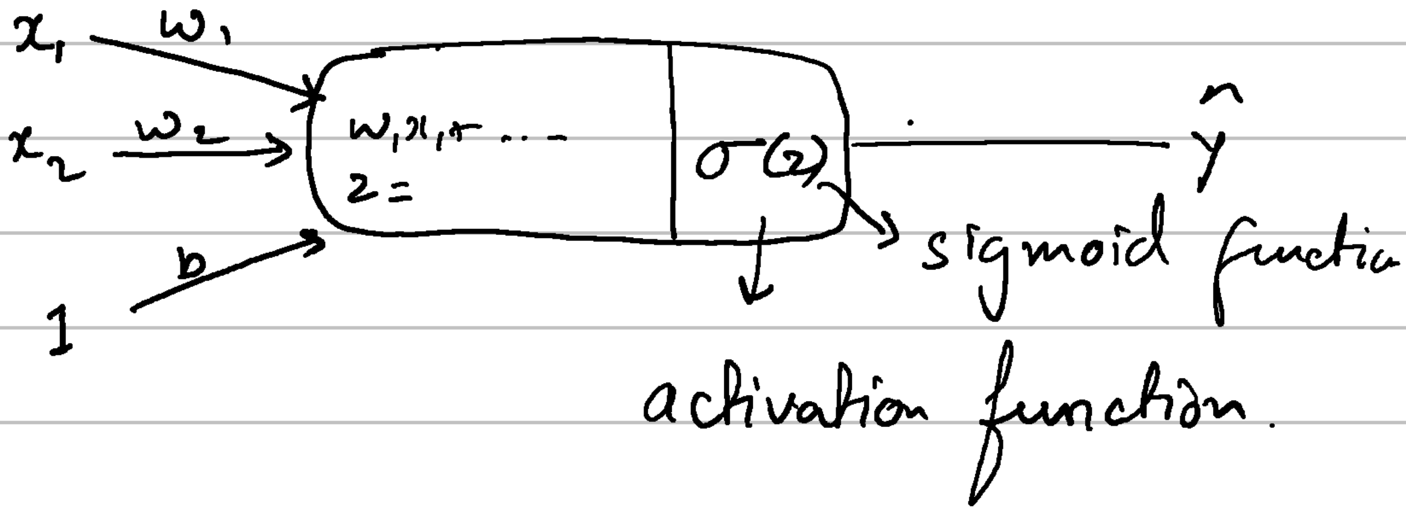
$$\frac{\partial L}{\partial w_2} = (w_1 x_1 + w_2 x_2 + b - y) \cdot w_2$$

# The data is usually normalized before forward / backprop.

$$x_{norm} = \frac{x - \hat{x}}{\sigma}$$

# Classification with Perceptron

## Activation function

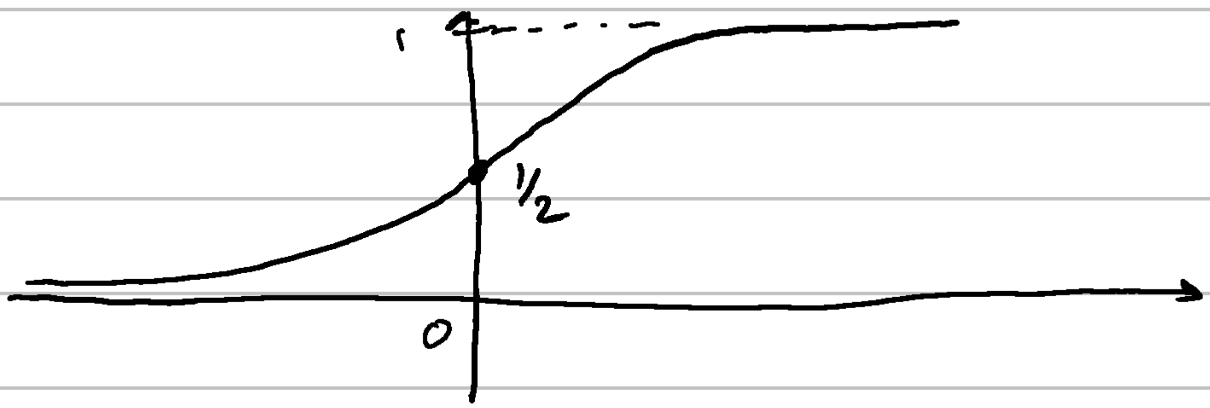



# Basically to convert z to something between $\underline{0 - 1}$ for classification

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

# Sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$




$$\sigma'(z) = -\frac{1}{(1+e^{-z})^2} \cdot (-1)\, e^{-z}$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\boxed{\sigma'(z) = \sigma(z) \cdot (1-\sigma(z))}$$

Loss function that works the best for classification problem is log-loss.

$$L(y, \hat{y}) = -y \ln(\hat{y}) - (1-y)\ln(1-\hat{y})$$

$L(y, \hat{y})$ — large when $y, \hat{y}$ are far

└ small when $y, \hat{y}$ are close.

\# Works better since it is a probabilistic function (classification problem)

$$\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + b)$$

$$L(y,\hat{y}) = -y\ln(\hat{y}) - (1-y)\ln(1-\hat{y})$$

$$\frac{\partial L}{\partial w_i} = -y \cdot \frac{1}{\hat{y}} \cdot \frac{d\hat{y}}{dw_i} - \frac{1-y}{1-\hat{y}} \cdot (-1)\frac{d\hat{y}}{dw_i}$$

$$= \frac{d\hat{y}}{dw_i}\left[\frac{\hat{y}(1-y) - (y)(1-\hat{y})}{(1-\hat{y})(\hat{y})}\right]$$

$$= \frac{d\hat{y}}{dw_i}\left[\frac{\hat{y} - \cancel{y\hat{y}} - y + \cancel{y\hat{y}}}{(1-\hat{y})(\hat{y})}\right]$$

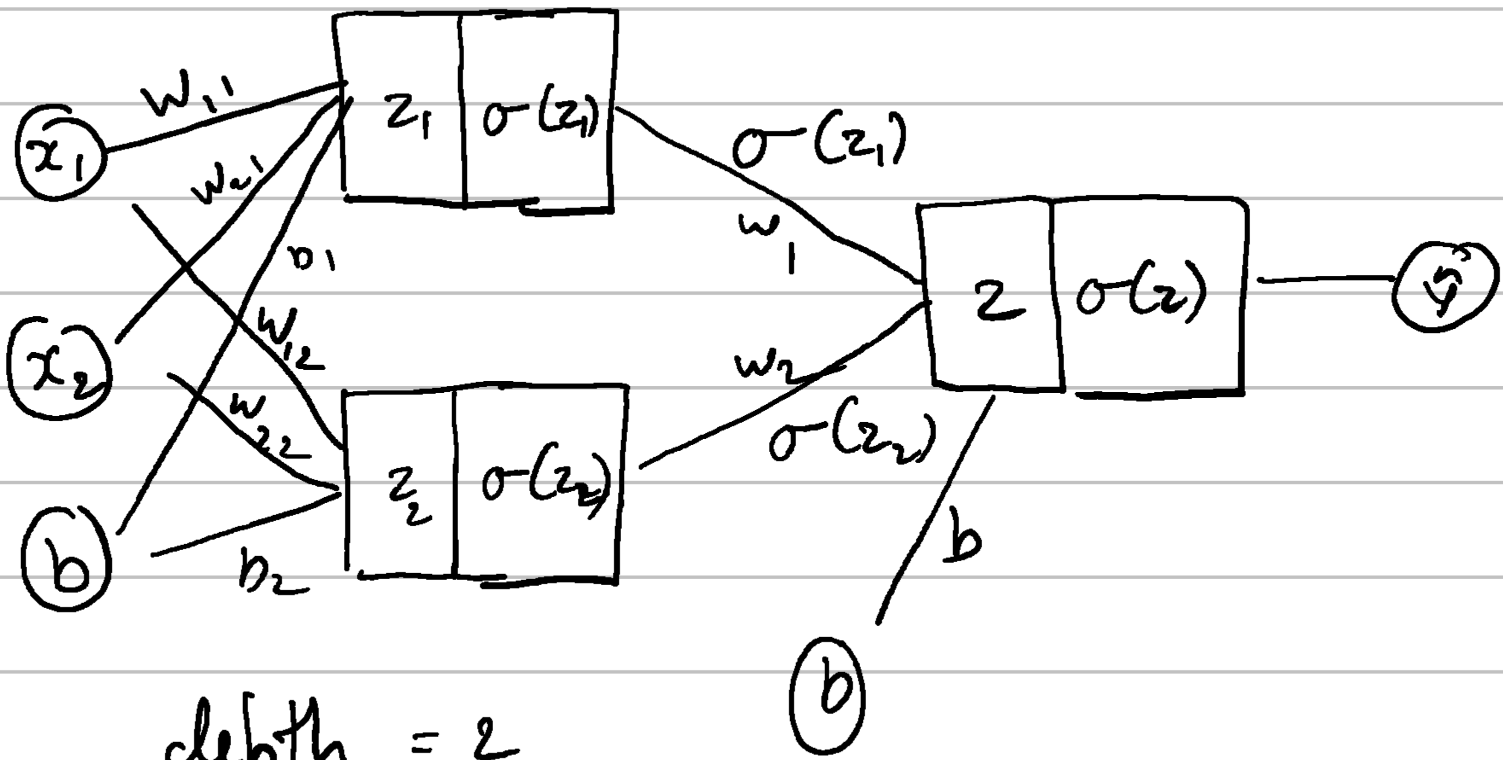
$$\frac{d\hat{y}}{dw_i} = x_i \cdot \hat{y}(1-\hat{y})$$

$$\frac{dL}{dw_i} = x_i\, \cancel{\hat{y}(1-\hat{y})} \cdot \left(\frac{\hat{y}-y}{\cancel{\hat{y}(1-\hat{y})}}\right)$$

$$\boxed{\frac{dL}{dw_i} = x_i \cdot (\hat{y}-y)}$$

# Classification with neural networks

⇒ A combination of perceptrons into various layers.




depth = 2

1 input layer
1 hidden layer
1 output layer.

$$\hat{y} = \sigma(w_1 a_1 + w_2 a_2 + b)$$

$$a_1 = \sigma(w_{11} x_1 + w_{21} x_2 + b_1)$$

$$a_2 = \sigma(w_{21} x_1 + w_{22} x_2 + b_2)$$

$$L(y, \hat{y}) = -y \ln(\hat{y}) - (1-y) \ln(1-\hat{y})$$

## Finding the derivatives

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_1} = a_1 \cdot (\hat{y} - y)$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_{22}} = a_2 \cdot (\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial b} = 1 \cdot (\hat{y} - y)$$

$$\hat{y} = \sigma(\underbrace{w_1 a_1 + w_2 a_2 + b}_{\hookrightarrow z})$$

$$a_1 = \sigma(\underbrace{w_{11} x_1 + w_{21} x_2 + b_1}_{\hookrightarrow z_1})$$

$$a_2 = \sigma(\underbrace{w_{21} x_1 + w_{22} x_2 + b_2}_{\hookrightarrow z_2})$$

$$\Rightarrow \frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial w_{11}} \rightarrow \frac{da_1}{dz_1} \cdot \frac{dz_1}{dw_{11}}$$

$$\boxed{\frac{d\hat{y}}{da_1} = \frac{d\hat{y}}{dz} \cdot \frac{dz}{da_1}}$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}}$$

$$= w_1 \cdot (\hat{y} - y) \cdot \sigma(z_1)(1 - \sigma(z_1)) \cdot x_1$$

$$= x_1 \cdot w_1 (\hat{y} - y) \cdot a_1 \cdot (1 - a_1)$$

# In a neural network we uptade the layers in order from input to output.